

1] Define cloud Computing and explain any four characteristics that make it different from traditional on-premise computing.

A] Cloud computing:-

Cloud computing is the delivery of computing servers, storage, databases and analytics over the internet. cloud computing enables business and individuals to use IT resources efficiently without investing in expensive infrastructure.

i] On-Demand self-service :-

User can create, modify or remove computing resources whenever required without interacting directly with the cloud Provider. This allows organization to quickly deploy virtual machine, storage whenever needed.

ii] Broad Network Access:-

Cloud services are accessible through the internet using standard devices such as desktops, laptops, and tablets. Users can access their applications and data from anywhere with an internet connection.

iii] Resource Pooling:-

Cloud Providers use virtualization technology to combine computing techniques into shared pool. Multiple customers share the same physical infrastructure.

iv] Rapid elasticity :-

cloud resources can automatically scale up or down depending on workload requirement. During high demand additional resources are allocated instantly and during low demand, unused resources are released.

Conclusion :-

cloud computing provides Flexibility, Scalability, cost savings and global accessibility. These characteristics make it a better alternative to traditional.

2] Trace the evolution of cloud computing from hardware evolution to internet software evolution and show how these changes enabled modern cloud platforms.

A] Cloud computing has evolved through several technological developments over many decades.

Initially, organizations used mainframe computers, where multiple users shared a single powerful computer through terminals. Although effective, these systems were expensive and available only to large organizations.

During the 1980s, personal computers (PCs) became popular, allowing individual users to own dedicated computer systems. However, organizations had to purchase and maintain separate

hardware for every user.

These development led to modern cloud Platforms that provide infrastructure as a services (IaaS), Platforms as a services (PaaS) and Software as a Service (SaaS). cloud providers now offer scalable and cost-effective computing resources through internet.

Conclusion:-

The evolution of hardware, virtualization, networking and web technologies laid the foundation for today's cloud computing Platforms, enabling organization to access computing resource anytime and anywhere.

3] Explain the role of server virtualization in cloud computing. differentiate between virtualization as an enabling technology and cloud computing as a service model.

A] server virtualization:-

Server virtualization is the process of creating multiple virtual servers on a single physical server using software called a hypervisor. each virtual machine operates independently with its own operating system and applications.

- The major roles of server virtualization
- Improving hardware utilization.
 - Reducing infrastructure and maintenance costs.
 - Supporting rapid development of virtual machines.
 - Enabling flexible resource allocation.

virtualization:

virtualization is an enabling technology that creates virtualized computing resources over the internet. It combines virtualization with automation, networking security.

→ cloud computing as a service model that delivers virtualized computing resources over the internet. It combines virtualization with automation and billing systems to provide services on demand.

virtualization	cloud computing
i] Creates virtual machines	i] Delivers computing services over the internet.
ii] Focuses on hardware utilization	ii] Focuses on service delivery
iii] Can be used without cloud	iii] uses virtualization as a foundation
iv] managed internally	iv] managed by cloud service providers.

Conclusion:

virtualization enables cloud computing by efficiently managing physical resources, while cloud computing uses virtualization to provide scalable and on demand IT services.

4) Discuss The role of data centers in delivering cloud services.

A data center is a secure facility containing servers, storage systems, networking equipment, cooling systems, power supplies, and security infrastructure.

The primary role of data center is to host computing resource such as virtual servers, database, storage system and networking equipment required by cloud customers.

Cloud Providers use data centers to achieve scalability by adding servers and storage whenever customer demand increases. virtualization work helps distribute workloads efficiently among available resources.

Conclusion:

Data centers provide secure, reliable, scalable and highly available infrastructure that enables cloud providers to deliver computing services efficiently to users around the world.

5] differentiate IaaS, PaaS, SaaS and XaaS with one suitable real-world example for each model.

A] Cloud computing provides different service models based on the level of responsibility shared between the cloud provider and the customer.
Infrastructure as a service (IaaS):

IaaS Provider virtualized computing infrastructure such as virtual machines, storage, networking and operating systems. A popular example is Amazon EC2.

Platform as a service (PaaS):

PaaS provides a complete development platform that allows developers to build, test and deploy application without managing hardware system.

Google App Engine is a common example.

Software as a service (SaaS):

SaaS delivers complete software applications over the internet. Users simply access the software through a web browser without installing or maintaining it. Gmail, Docs, Drive is a common example.

Anything as a service (XaaS):

XaaS refers to delivering any IT service through the cloud, including storage, database, cloud storage services is a suitable example of XaaS.

Conclusion:-

Organization select the appropriate model based on their business needs, technical expertise and budget.